

XIAOYANG WAN

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Education

University of Pennsylvania

M.S. in Computer and Information Science

Sep. 2025 – May 2027

Philadelphia, PA

Zhejiang University

B.S. in Computer Science and Technology; GPA: 3.96/4.00

Sep. 2021 – May 2025

Zhejiang, China

Professional Experience

Upstart

Jun. 2026 – Aug. 2026

Machine Learning Engineer Intern

San Jose, CA

- Scaled BERT embeddings feeding the personal-loan underwriting model across **16 single-GPU** Metaflow shards with indexed **Parquet** reads; validation runs: **2x** faster, **80%** cheaper, **90%** less peak RAM.
- Traced bin flips in **2.13%** of borrower records to a 1-ULP **NumPy log1p** drift across hosts; froze raw-amount thresholds to restore **1e-5** embedding parity and remove a source of train/serve skew.
- Found CPU tokenization starving GPU inference and vectorized it, lifting GPU utilization $\sim$ **4x**.
- Benchmarked GPU training of a six-bag **PyTorch** ensemble with GPU-resident feature matrices and parallel feature prep; verified **1e-5** CPU/GPU output parity and cut runtime **3h21m to 1h38m** and cost **73%**.
- Shipped fixes for **AWS Batch** GPU contention and **16-min** TCP stalls in the company-wide **@batch** decorator.

Molardata & 2077 AI

Apr. 2025 – Aug. 2025

Machine Learning Engineer Intern

Zhejiang, China

- Built a **LangChain** multi-agent pipeline (**GPT-4o** generator, **GPT-4o-mini** critic for missed attack types, GPT-4o coverage judge) that produced validated adversarial tests for **1,700** Codeforces problems.
- Gated LLM verdicts behind execution checks (input verifier, reference and known-wrong solutions), reaching near-**100%** error detection in a pilot; ran up to three feedback rounds with per-step **checkpoints**.
- Built an **LLM-as-judge** for **200** three-turn **FLUX** image-editing sequences, comparing result and target captions; **83%** passed manual review, vs. an embedding baseline that caught only half of identity drifts.

DCD Lab, Zhejiang University

Jun. 2024 – Jan. 2025

Research Assistant

Zhejiang, China

- Adapted and ran seven image-editing data pipelines, producing **500K+** training pairs for AnyEdit (**CVPR 2025 Oral**); evaluated edits with **CLIP**, **DINO**, and L1 metrics.
- Sharded generation across two GPUs with skip-if-exists **checkpointing** for resumable batches.

Selected Projects

Flight Delay Prediction | CIS 5450 Big Data Analytics, Team of 4; owned modeling

Apr. 2026

- Modeled 15+ minute delays on **6.8M** 2024 U.S. flights joined with NOAA hourly weather, progressing from logistic regression and random forest to **XGBoost** and **LightGBM** under a time-based split.
- Tuned XGBoost to **0.817 AUC**; class weights beat SMOTE on recall and AUC, a tuned threshold gave 0.663 precision at 0.525 recall, and AUC held within **0.002** on the full 5.6M-row set.
- Tested delay drivers with permutation tests (budget vs. legacy carriers, hub vs. non-hub), bootstrap CIs (summer vs. winter), and a Monte Carlo χ^2 test for weather.

QUAKER: Two-Stage Product Retrieval | CIS 5200 Machine Learning Course Project

Dec. 2025

- Fine-tuned a **BERT-large** category classifier ($\sim$ 90% top-4 hit rate) to filter items before **E5** ranking on 21K vague queries; reached **74.8%** top-200 accuracy, **+32** pts over TF-IDF.
- Traced a batched-vs-single scoring mismatch to per-batch padding lengths and fixed it with fixed-length padding; precomputed item embeddings to make evaluation $\sim$ **45x** faster.

Publication

CVPR 2025 Oral | *AnyEdit: Mastering Unified High-Quality Image Editing for Any Idea* | Co-author.

Technical Skills

Statistics & Modeling: Hypothesis Testing, Supervised Learning, XGBoost, LightGBM, Class Imbalance, Model Validation

Machine Learning: PyTorch, scikit-learn, BERT, E5, Embeddings, Feature Engineering, Ensembles, LLM-as-Judge

Data: Python, SQL, NumPy, pandas, Parquet/PyArrow, Large-Scale Batch Processing, Vectorization

LLM & Vision: LangChain, Multi-Agent Systems, Prompt Engineering, CLIP, Diffusion Models

Platforms: Metaflow, AWS Batch, EC2, S3, Docker, MLflow, GPU Training & Inference, Linux, Git